Project Outline: Interactive Storytelling Platform for Pilgrim Protect

Project Overview

Pilgrim Protect is an initiative by Pilgrim Africa aimed at combating malaria in high-burden areas, starting with Uganda, while fostering long-term local capacity. The program begins by providing free insecticide spraying for schools, transitioning to partial or full cost recovery in subsequent years. Funds from donors and partial payments enable expansion to more schools, creating a sustainable model. Success is measured by scaling impact: for example, a donor sponsoring one school in year one could enable funding for two schools in year two through efficient resource allocation.

This project tasks a group of interns or students with building a flexible, interactive web-based platform (e.g., on protect.pilgrimafrica.org) to tell Pilgrim Protect's story. The platform will engage donors, highlight progress, and facilitate sponsorships. It emphasizes visual storytelling through maps, images, videos, and reports, while integrating tools for sharing and updates. The solution should be modular, allowing easy additions like email campaigns or expansions to other African countries.

The project is ideal for a team of 4-6 interns/students with skills in web development, design, data visualization, and content creation. It can be completed in 8-12 weeks, using open-source tools for cost-effectiveness.

Objectives

- Educate and Engage Users: Create an immersive experience showcasing malaria's impact, Pilgrim Protect's interventions, and success stories to inspire action.
- Facilitate Sponsorships: Enable donors to explore, select, and sponsor schools, with real-time updates on spraying progress and outcomes.
- Promote Sustainability: Illustrate the program's self-funding model through interactive visuals, demonstrating how donations multiply impact over time.
- Build Capacity: Develop a scalable platform that Pilgrim Africa staff can maintain and expand, incorporating user feedback for future features like email campaigns.

Scope and Features

The platform will be a responsive web app, potentially built on frameworks like React.js for the frontend, Node.js/Express for the backend, and a database like MongoDB or Firebase for data storage. Key features include:

1. Interactive Map:
    - Display high-burden districts in Uganda (expandable to other African countries like Kenya or Tanzania).
    - Use libraries like Leaflet or Google Maps API to plot schools: green markers for protected schools, red for those needing sponsors.
    - Users can zoom, filter by district or burden level, and click markers for details (e.g., school name, student count, malaria incidence data).
    - Integrate public health data sources (e.g., WHO APIs) for real-time malaria burden stats.

2. School Profiles and Media Gallery:
    - Detailed pages for each school with photos of facilities, spraying events, and before/after comparisons.
    - Embed videos from Pilgrim workers sharing on-the-ground stories (e.g., via YouTube or Vimeo integration).
    - Include progress reports: charts showing malaria case reductions (e.g., using Chart.js), testimonials from teachers/students, and impact metrics.

3. Donor Tools:
    - Sponsorship portal: Users browse schools, view costs (e.g., $X for one spraying cycle), and sponsor via integrated payment gateway (e.g., Stripe or PayPal).
    - Sharing features: Social media buttons to share school profiles or success stories.
    - Personalized updates: Donors receive email notifications (via tools like Mailchimp) when spraying occurs, with photos and reports. Envision an automated email campaign series: welcome, progress updates, renewal reminders.

4. Storytelling Elements:
    - Narrative sections explaining the model: Interactive timeline showing how one sponsorship grows (e.g., Year 1: 1 school; Year 2: 2 schools via partial local funding).
    - Blog-style updates from field teams, with options for user comments or Q&A.
    - Accessibility features: Mobile-friendly design, alt text for images, and multilingual support (English, local Ugandan languages).

Out-of-scope: Full e-commerce setup (focus on prototype integration); advanced AI analytics; physical hardware.

Technical Requirements

- Frontend: HTML/CSS/JavaScript with React/Vue for interactivity; responsive design via Bootstrap or Tailwind CSS.
- Backend: Node.js for API endpoints (e.g., fetching school data); authentication for donor accounts.
- Data Management: Store school info, donor details, and media in a cloud database; use AWS S3 or similar for image/video hosting.
- Security: Implement HTTPS, data encryption, and basic user privacy (GDPR-compliant).
- Tools and Skills Needed: Git for version control; basic UI/UX design (Figma for prototypes); content sourcing from Pilgrim Africa team.

Timeline (8-Week Sprint)

- Weeks 1-2: Research and Planning – Gather requirements, design wireframes, source initial data (e.g., school lists, photos).
- Weeks 3-4: Development – Build core map and school profiles; integrate media.
- Weeks 5-6: Add Interactivity – Implement sponsorship tools, sharing, and basic email mockups.
- Weeks 7-8: Testing and Polish – User testing, bug fixes, documentation for handover; demo to Pilgrim Africa.

Team Roles

- Project Lead: Coordinates tasks, liaises with Pilgrim Africa.
- Developer(s): Handles coding for map, backend, and integrations (2-3 members).
- Designer/Content Creator: Creates visuals, writes narratives, sources media.
- Tester/Analyst: Ensures functionality, gathers feedback, measures usability.

Success Metrics and Deliverables

- Quantitative: Platform prototype with at least 10 sample schools; 80%+ user satisfaction in testing; simulated sponsorship flow.
- Qualitative: Engaging storytelling that clearly conveys the scaling model; flexible codebase for future additions (e.g., email campaigns).
- Deliverables: Deployed demo site, source code repository, user guide, presentation deck.

This project not only advances Pilgrim Protect's mission but also provides hands-on experience in impactful tech for good. Estimated budget: Minimal (free tools), with potential for Pilgrim Africa to provide data and feedback